THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Rust is formed on: 1

(a) iron **(b)** silver **(c)** copper **(d)** gold

2. A triple bond is present in: 1

(a) ethyne **(b)** ethane **(c)** methane **(d)** ethene

3. The colour of lead iodide precipitate is: 1

(a) yellow **(b)** red **(c)** white **(d)** blue

4. Acid + metal carbonate gives salt, water and: 1

(a) hydrogen **(b)** nitrogen **(c)** carbon dioxide **(d)** oxygen

5. The ore of copper is: 1

(a) cinnabar **(b)** bauxite **(c)** copper glance **(d)** galena

6. Which is the hardest substance? 1

(a) diamond **(b)** fullerene **(c)** coal **(d)** graphite

7. The chemical formula of slaked lime is: 1

(a) $CaSO_4$ **(b)** $CaCO_3$ **(c)** CaO **(d)** $Ca(OH)_2$

8. Red litmus turns blue in: 1

(a) acid **(b)** base **(c)** vinegar **(d)** lemon juice

9. Metals that produce a ringing sound when struck are called: 1

(a) malleable **(b)** brittle **(c)** sonorous **(d)** ductile

10. The formula of methane is: 1

(a) C_2H_6 **(b)** C_3H_8 **(c)** CH_4 **(d)** C_2H_4

11. $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ is a: 1

(a) double displacement reaction **(b)** decomposition reaction
(c) displacement reaction **(d)** combination reaction

12. Universal indicator is used to find: 1

(a) the approximate pH **(b)** the density
(c) the temperature **(d)** the mass

13. The formula of rust is: 1

(a) FeS **(b)** $FeCl_3$ **(c)** $Fe_2O_3 \cdot x\,H_2O$ **(d)** FeO

14. Alkanes are: 1

(a) alcohols **(b)** saturated hydrocarbons
(c) acids **(d)** unsaturated hydrocarbons

15. When an iron nail is placed in copper sulphate solution, the deposit on the nail is: 1

(a) reddish-brown **(b)** green **(c)** white **(d)** black

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Name the products when magnesium burns in oxygen. 2

17. Write the general formula of alkane and alkene. 2

18. Write the word equation for burning of magnesium in oxygen. 2

19. Define a base and give two examples. 2

20. Name two metals that are good conductors of electricity. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Write any three physical properties of non-metals. 3

22. Write the names of the first three alkenes with formulas. 3

23. Name the type of reaction: $CaCO_3 \longrightarrow CaO + CO_2$. Give a reason. 3

24. State three properties of bases. 3

OR

Explain with an equation how magnesium oxide reacts with water. 3

25. State three uses of ethanol. 3

OR

Magnesium ribbon is burnt in air. State three observations. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Lead nitrate is heated in a test tube. Write the observation, equation and type of reaction. Name the brown gas. 5

OR

Write equations for the reaction of dilute HCl with 5

(a) zinc
(b) NaOH
(c) Na_2CO_3
(d) CuO and
(e) $NaHCO_3$.

27. Describe what happens when sodium, magnesium and iron react with water. Write equations. 5

OR

Differentiate between saturated and unsaturated hydrocarbons (five points). 5

END OF QUESTION PAPER **Total: 50 Marks**

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